Nonlinear field equations and FLRW cosmology of the spectral action with Standard Model content

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Abstract

We derive the full nonlinear variational field equations of the one-loop spectral action $S = \mathrm{Tr}\, f(D^2/\Lambda^2)$ on arbitrary curved backgrounds and reduce them to Friedmann–Lemaître–Robertson–Walker (FLRW) cosmology. The spectral action with Standard Model content produces a nonlocal effective action in the curvature-squared sector with two entire form factors $F_1(\Box/\Lambda^2)$ (Weyl-squared) and $F_2(\Box/\Lambda^2, \xi)$ (Ricci-scalar-squared), whose local limits yield the parameter-free coefficient $\alpha_C = 13/120$ and the ξ-dependent coefficient $\alpha_R(\xi) = 2(\xi - 1/6)^2$. The variation $\delta S/\delta g^{\mu\nu} = 0$ gives the field equations

$$\frac{1}{\kappa^2}\, G_{\mu\nu} + \frac{1}{16\pi^2}\big[2\, F_1\, B_{\mu\nu} + \Theta^{(C)}_{\mu\nu}\big] + \frac{1}{16\pi^2}\big[F_2\, H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\big] = \tfrac{1}{2}\, T_{\mu\nu},$$

where $B_{\mu\nu}$ is the nonlocal Bach tensor, $H_{\mu\nu}$ arises from the R^2 variation, and $\Theta^{(C,R)}_{\mu\nu}$ are Barvinsky–Vilkovisky alpha-insertion corrections from the metric dependence of $\Box$. On FLRW backgrounds, the entire Weyl sector vanishes by conformal flatness, and only the R^2 sector contributes to the modified Friedmann equation. We establish five central results: (i) the gravitational-wave speed satisfies $c_T = c$ to $\mathcal{O}(H^2/\Lambda^2)$, passing the GW170817 constraint by 46 orders of magnitude; (ii) de Sitter spacetime is an exact, stable solution; (iii) at conformal coupling $\xi = 1/6$, all spectral corrections vanish identically; (iv) late-time corrections to the Hubble parameter are $\delta H^2/H^2 \sim 10^{-64}$, ensuring automatic consistency with all precision cosmological data; (v) the local limit recovers Stelle's higher-derivative gravity with calculable coefficients. All results are verified by dual independent derivations and confirmed by numerical evaluation at 100-digit precision.

Keywords: spectral action, nonlocal gravity, field equations, FLRW cosmology, gravitational-wave speed, Standard Model

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1 Introduction

The spectral action principle [1, 2] provides a geometric origin for both gravitational and gauge interactions: the bosonic action is given by $S = \mathrm{Tr}\, f(D^2/\Lambda^2)$, where D is the Dirac operator of a noncommutative geometry, Λ is a cutoff scale, and f is a positive even function. At the classical level, the heat kernel expansion reproduces the Einstein–Hilbert action, cosmological constant, and Yang–Mills terms from the leading Seeley–DeWitt coefficients [3, 4].

Beyond the leading order, the spectral action generates *nonlocal* curvature-squared corrections characterized by momentum-dependent form factors. In a companion paper [5] (hereafter Paper I), we computed the complete one-loop form factors $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2, \xi)$ for the full Standard Model particle content: 4 real scalars (Higgs doublet), 45/2 Dirac-equivalent fermions (3 generations), and 12 gauge bosons (SU(3) $\times$ SU(2) $\times$ U(1)), following the Barvinsky–Vilkovisky covariant perturbation theory [6, 7] and the Codello–Zanusso diagrammatic heat kernel [8]. Both form factors are entire functions of $\Box/\Lambda^2$, ensuring that the one-loop effective action introduces no additional propagator poles beyond those of the classical theory. In a second companion paper [9] (Paper II), we extracted the linearized field equations on a flat background, derived the modified graviton propagator and Newtonian potential, and confronted the predictions with solar-system and laboratory experiments.

The present paper completes the program by deriving the *full nonlinear* field equations on arbitrary curved backgrounds and reducing them to FLRW cosmology. Whereas the linearized analysis of Paper II is sufficient for weak-field phenomenology, astrophysical and cosmological applications require the complete variational equations, including the Barvinsky–Vilkovisky alpha-insertion corrections that arise from the nontrivial metric dependence of the d'Alembertian $\Box$.

The paper is organized as follows. Section 2 fixes notation and conventions. Section 3 reviews the spectral action and its curvature-squared sector. Section 4 presents the variational calculus for nonlocal operators. Section 5 states and proves the full nonlinear field equations. Section 6 verifies their fundamental properties: local limit, linearized limit, Bianchi identity, trace structure, de Sitter solutions, and Ricci-flat simplification. Section 7 briefly reviews the linearized limit and the modified Newtonian potential (detailed in Paper II). Section 8 reduces the field equations to FLRW cosmology, deriving the modified Friedmann equations. Section 9 establishes the gravitational-wave speed. Section 10 treats de Sitter stability. Section 11 quantifies the late-time cosmological suppression. Section 12 compares with Stelle gravity, $f(R)$ theories, and nonlocal gravity

models. Section 13 concludes. Appendix A collects form factor identities used in the derivations.

2 Notation and conventions

We work in $d = 4$ dimensions. The full nonlinear field equations are derived in Euclidean signature $(+,+,+,+)$, while the FLRW reduction uses Lorentzian signature $(-,+,+,+)$; the Wick rotation connecting the two is detailed in Paper I. The conventions are:

Symbol	Definition
$g_{\mu\nu}$	Metric tensor
$R^{\rho}{}_{\sigma\mu\nu}$	Riemann tensor: $\partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \cdots$
$R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$	Ricci tensor
$R = g^{\mu\nu} R_{\mu\nu}$	Ricci scalar
$C_{\mu\nu\rho\sigma}$	Weyl tensor
$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$	Einstein tensor
$\Box = g^{\mu\nu} \nabla_\mu \nabla_\nu$	Covariant d'Alembertian
$\kappa^2 = 8\pi G$	Gravitational coupling
Λ	Spectral cutoff scale
ξ	Higgs non-minimal coupling to gravity

The Weyl tensor satisfies the decomposition

$$C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 2\, R_{\mu\nu} R^{\mu\nu} + \tfrac{1}{3}\, R^2, \tag{1}$$

and the Gauss–Bonnet topological density is

$$E_4 = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 4\, R_{\mu\nu} R^{\mu\nu} + R^2. \tag{2}$$

On a closed four-manifold, $\int \mathrm{d}^4x \sqrt{g}\, E_4 = 32\pi^2 \chi(M)$.

3 The spectral action and its curvature-squared sector

3.1 One-loop effective action

From the spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ and the Seeley–DeWitt heat kernel expansion, the curvature-squared sector of the one-loop effective action is (Paper I)

$$S_{\mathrm{quad}} = \frac{1}{16\pi^2} \int \mathrm{d}^4x \sqrt{g} \left[F_1(\Box/\Lambda^2)\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} + F_2(\Box/\Lambda^2, \xi)\, R^2 \right]. \tag{3}$$

The $\{C^2, R^2\}$ basis is chosen because (i) it separates the spin-2 (transverse-traceless) and spin-0 (scalar) gravitational sectors at the linearized level; (ii) the Gauss–Bonnet identity (2) removes one degree of freedom from the three-invariant basis $\{R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}, R_{\mu\nu} R^{\mu\nu}, R^2\}$; and (iii) the form factors F_1, F_2 are entire functions of $\Box/\Lambda^2$ (Paper I), ensuring no spurious poles.

The total action including the Einstein–Hilbert term and matter is

$$S[g] = \frac{1}{2\kappa^2} \int \mathrm{d}^4x \sqrt{g}\, R + S_{\mathrm{quad}}[g] + S_{\mathrm{matter}}[g]. \tag{4}$$

3.2 Standard Model coefficients

The SM-summed local limits are (Paper I):

$$\alpha_C \equiv 16\pi^2\, F_1(0) = \frac{13}{120}, \tag{5}$$

$$\alpha_R(\xi) \equiv 16\pi^2\, F_2(0,\xi) = 2\Big(\xi - \frac{1}{6}\Big)^2, \tag{6}$$

with SM multiplicities $N_s = 4$, $N_D = 45/2$, $N_v = 12$ following Codello–Percacci–Rahmede [10]. The normalized form factors $\hat{F}_i(z) = F_i(z)/F_i(0)$ satisfy $\hat{F}_i(0) = 1$ and are built from the master function

$$\varphi(x) = e^{-x/4}\sqrt{\frac{\pi}{x}}\,\mathrm{erfi}\Big(\frac{\sqrt{x}}{2}\Big), \quad \varphi(0) = 1, \;\; \varphi'(0) = -\frac{1}{6}. \tag{7}$$

The ratio in the $\{R^2, R^2_{\mu\nu}\}$ basis is

$$\frac{c_1}{c_2} = -\frac{1}{3} + \frac{120(\xi - 1/6)^2}{13}, \tag{8}$$

which equals $-1/3$ at conformal coupling $\xi = 1/6$.

4 Variational calculus for nonlocal operators

4.1 The central difficulty

The variation of the nonlocal action (3) with respect to $g^{\mu\nu}$ differs from the local case because the d'Alembertian $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ itself depends on the metric. Consequently,

$$\frac{\delta}{\delta g^{\mu\nu}}\big[F(\Box)\big] \neq 0, \tag{9}$$

and the variation of $A\,F(\Box)\,B$ receives an additional contribution from $\delta F(\Box)/\delta g^{\mu\nu}$, beyond the "naive" terms $(\delta A)\,F(\Box)\,B + A\,F(\Box)\,(\delta B)$.

4.2 The Barvinsky–Vilkovisky alpha-insertion formula

The systematic treatment of this problem was given by Barvinsky and Vilkovisky [6, 11]. For the variation of $\int \mathrm{d}^4x\,\sqrt{g}\,A\,F(\Box)\,B$:

$$\frac{\delta}{\delta g^{\mu\nu}}\big[A\,F(\Box)\,B\big] = (\delta A)\,F(\Box)\,B + A\,F(\Box)\,(\delta B) + \int_0^1 \mathrm{d}\alpha\,[e^{\alpha\Box}\,A]\,\frac{\delta F(\Box)}{\delta g^{\mu\nu}}\,[e^{(1-\alpha)\Box}\,B]. \tag{10}$$

The third term, the alpha-insertion, generates the correction tensors $\Theta^{(C)}_{\mu\nu}$ and $\Theta^{(R)}_{\mu\nu}$.

4.3 Taylor expansion form

Since $F(\Box)$ is entire, we expand $F(z) = \sum_{n=0}^{\infty} c_n z^n$ and write the alpha-insertion for scalar quantities A, B as

$$\Theta_{\mu\nu} = -\sum_{n=1}^{\infty} c_n \sum_{k=0}^{n-1}\Big[\nabla_\mu(\Box^k A)\,\nabla_\nu(\Box^{n-1-k}B) - \tfrac{1}{2}\,g_{\mu\nu}\,\nabla_\rho(\Box^k A)\,\nabla^\rho(\Box^{n-1-k}B)\Big]. \tag{11}$$

This is the Leibniz-rule expansion for the variation of $\Box^n$, valid when A and B are scalars.

Proposition 4.1 (Properties of $\Theta_{\mu\nu}$)**.** *The alpha-insertion tensor satisfies:*
(i) ***Local limit:*** *If $F = \text{const}$ (i.e., $c_n = 0$ for $n \geq 1$), then $\Theta_{\mu\nu} = 0$.*
(ii) ***De Sitter:*** *If $A = B = R = \text{const}$, then $\nabla R = 0$ and $\Theta_{\mu\nu} = 0$.*
(iii) ***Linearized on flat:*** *On a flat background, $\Theta_{\mu\nu} = \mathcal{O}(h^2)$, so it does not contribute to the linearized propagator.*
(iv) ***Symmetry:*** *$\Theta_{\mu\nu} = \Theta_{\nu\mu}$.*
(v) ***Bianchi identity:*** *$\nabla^\mu \Theta_{\mu\nu} = 0$, as a consequence of the diffeomorphism invariance of the full action.*

5 Full nonlinear field equations

5.1 The R^2 sector

The variation of $\sqrt{g}\, R\, F_2(\Box)\, R$ with respect to $g^{\mu\nu}$ yields, in the "naive" part, the tensor

Definition 5.1 (The H tensor)**.**

$$H_{\mu\nu} \coloneqq 2\nabla_\mu \nabla_\nu R - 2\, g_{\mu\nu} \Box R - \tfrac{1}{2}\, g_{\mu\nu}\, R^2 + 2R\, R_{\mu\nu}. \tag{12}$$

Its trace is

$$g^{\mu\nu} H_{\mu\nu} = -6\, \Box R, \tag{13}$$

and it vanishes on maximally symmetric spacetimes: $H_{\mu\nu}|_{\text{dS}} = 0$.

The alpha-insertion for this sector, with $A = B = R$, is given by (11) with the Taylor coefficients of F_2:

$$\Theta^{(R)}_{\mu\nu} = -\sum_{n=1}^{\infty} c_n^{(2)} \sum_{k=0}^{n-1} \Big[\nabla_\mu (\Box^k R)\, \nabla_\nu (\Box^{n-1-k} R) - \frac{1}{2}\, g_{\mu\nu}\, \nabla_\rho (\Box^k R)\, \nabla^\rho (\Box^{n-1-k} R) \Big], \tag{14}$$

where $F_2(z, \xi) = \sum_{n=0}^{\infty} c_n^{(2)}\, (z/\Lambda^2)^n$.

5.2 The Weyl C^2 sector

The variation of $\sqrt{g}\, C_{\alpha\beta\gamma\delta}\, F_1(\Box)\, C^{\alpha\beta\gamma\delta}$ yields the nonlocal Bach tensor:

Definition 5.2 (Bach tensor)**.**

$$B_{\mu\nu} = \nabla^\rho \nabla^\sigma C_{\mu\rho\nu\sigma} + \tfrac{1}{2}\, R^{\rho\sigma} C_{\mu\rho\nu\sigma}. \tag{15}$$

The Bach tensor is symmetric, traceless ($g^{\mu\nu} B_{\mu\nu} = 0$ from the conformal invariance of $\int \sqrt{g}\, C^2$ in four dimensions [12]), and divergence-free ($\nabla^\mu B_{\mu\nu} = 0$).

The alpha-insertion for the Weyl sector, $\Theta^{(C)}_{\mu\nu}$, has a more involved structure than (14) because it acts on rank-4 Weyl tensors, and the commutator $[\Box, \nabla_\mu]$ on tensors generates Riemann-curvature insertions at each order. The key property is that $\Theta^{(C)}_{\mu\nu}$ vanishes in all the limiting cases relevant for cosmology: on flat backgrounds, on de Sitter ($C_{\mu\nu\rho\sigma} = 0$), and at the linearized level ($\mathcal{O}(h^2)$).

5.3 The complete field equations

Combining the Einstein–Hilbert, Weyl, and R^2 sectors:

Theorem 5.3 (SCT nonlinear field equations)**.** *The field equations $\delta S/\delta g^{\mu\nu} = 0$ are*

$$\boxed{\frac{G_{\mu\nu}}{\kappa^2} + \frac{2F_1 B_{\mu\nu} + \Theta^{(C)}_{\mu\nu} + F_2 H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}}{16\pi^2} = \frac{T_{\mu\nu}}{2}} \tag{16}$$

where $F_1 = F_1(\Box/\Lambda^2)$ and $F_2 = F_2(\Box/\Lambda^2, \xi)$. More explicitly,

$$\begin{aligned}\frac{1}{\kappa^2}\, G_{\mu\nu} &+ \frac{1}{16\pi^2}\Big[2\, F_1\Big(\frac{\Box}{\Lambda^2}\Big) B_{\mu\nu} + \Theta^{(C)}_{\mu\nu}\Big] \\ &+ \frac{1}{16\pi^2}\Big[F_2\Big(\frac{\Box}{\Lambda^2}, \xi\Big) H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\Big] = \frac{1}{2}\, T_{\mu\nu},\end{aligned} \tag{17}$$

where $G_{\mu\nu}$ is the Einstein tensor, $B_{\mu\nu}$ is the Bach tensor (15), $H_{\mu\nu}$ is the R^2-sector tensor (12), $\Theta^{(C,R)}_{\mu\nu}$ are the Barvinsky–Vilkovisky alpha-insertion corrections, and $T_{\mu\nu}$ is the matter stress-energy tensor.

Proof. The Einstein–Hilbert variation is standard: $\delta S_{\mathrm{EH}}/\delta g^{\mu\nu} = -(2\kappa^2)^{-1}\, G_{\mu\nu}$. The curvature-squared variation is decomposed as

$$\frac{\delta S_{\mathrm{quad}}}{\delta g^{\mu\nu}} = \frac{1}{16\pi^2}\big[2\, F_1(0)\, B_{\mu\nu} + F_2(0,\xi)\, H_{\mu\nu} + \Theta^{(C)}_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\big] + \mathcal{O}(\Box F_i), \tag{18}$$

where the $\mathcal{O}(\Box F_i)$ terms are included in the full form factors acting on $B_{\mu\nu}$ and $H_{\mu\nu}$ respectively. The factor of 2 in front of $B_{\mu\nu}$ arises because $\delta(C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma})/\delta g^{\mu\nu} = 2B^{\mu\nu}$ (the Weyl tensor appears quadratically).

As an independent cross-check, we re-derived the field equations using the $\{R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, R_{\mu\nu}R^{\mu\nu}, R^2\}$ basis with coefficients $\gamma_4 = F_1/(16\pi^2)$, $\gamma_2 = -2F_1/(16\pi^2)$, $\gamma_0 = (F_2 + F_1/3)/(16\pi^2)$, then converted via Gauss–Bonnet. The condition $2\gamma_4 + \gamma_2 = 0$ ensures that the $R_{\mu\nu}R^{\mu\nu}$ sector cancels, and the combined result is identical to (16). □

6 Properties of the field equations

6.1 Local limit: recovery of Stelle gravity

In the local limit $F_i \to F_i(0) = \text{const}$, all alpha-insertion terms vanish ($\Theta^{(C,R)}_{\mu\nu} = 0$ by Proposition 4.1(i)), and the field equations reduce to

$$G_{\mu\nu} + 2\alpha\, B_{\mu\nu} + \beta\, H_{\mu\nu} = \kappa^2\, T_{\mu\nu}, \tag{19}$$

where

$$\alpha = \frac{\alpha_C}{16\pi^2} = \frac{13/120}{16\pi^2}, \qquad \beta = \frac{\alpha_R(\xi)}{16\pi^2} = \frac{2(\xi - 1/6)^2}{16\pi^2}. \tag{20}$$

This is precisely the Stelle higher-derivative gravity field equation [12], with calculable SM coefficients.

6.2 Linearized limit

On a flat background $g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}$, all Θ terms are $\mathcal{O}(h^2)$ (Proposition 4.1(iii)) and drop out at linear order. The "naive" variation terms reproduce the Barnes–Rivers [13, 14] projector structure with propagator denominators

$$\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z), \qquad \Pi_{\mathrm{s}}(z,\xi) = 1 + 6\Big(\xi - \frac{1}{6}\Big)^2 z\, \hat{F}_2(z,\xi), \tag{21}$$

where $z = k^2/\Lambda^2$. This is verified against the independent derivation in Paper II.

6.3 Bianchi identity

Diffeomorphism invariance of the total action implies

$$\nabla^\mu \left[\frac{1}{\kappa^2}\, G_{\mu\nu} + \frac{1}{16\pi^2}\big(2F_1\, B_{\mu\nu} + \Theta^{(C)}_{\mu\nu} + F_2\, H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\big) \right] = 0. \tag{22}$$

At the linearized level, this reduces to the transversality of the Barnes–Rivers projectors: $k^\mu P^{(2)}_{\mu\nu\rho\sigma} = 0$. The identity has been verified numerically on 20 random momenta with residual below 10^{-16}.

6.4 Trace structure

The trace of (16) gives

$$-\frac{R}{\kappa^2} - \frac{6}{16\pi^2}\, F_2\, \Box R + g^{\mu\nu}\big(\Theta^{(C)}_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\big) = \frac{1}{2}\, T, \tag{23}$$

using $g^{\mu\nu} G_{\mu\nu} = -R$, $g^{\mu\nu} B_{\mu\nu} = 0$ (conformal invariance of C^2), and $g^{\mu\nu} H_{\mu\nu} = -6\,\Box R$. In the local limit, this becomes $-R - 6\beta\,\Box R = \kappa^2 T$, consistent with the Stelle trace equation [15].

6.5 De Sitter exact solution

On a maximally symmetric spacetime with constant curvature R_0: $C_{\mu\nu\rho\sigma} = 0$ (Weyl-flat), $R = R_0 = \text{const}$ implies $H_{\mu\nu}|_{\mathrm{dS}} = 0$, and all alpha-insertion terms vanish (Proposition 4.1(ii)). The field equations reduce to pure Einstein gravity with effective cosmological constant $\Lambda_{\mathrm{eff}} = R_0/4$.

6.6 Ricci-flat simplification

On Ricci-flat backgrounds ($R = R_{\mu\nu} = 0$; e.g., Schwarzschild, Kerr), $H_{\mu\nu} = 0$ and $\Theta^{(R)}_{\mu\nu} = 0$. The Weyl tensor is generally nonzero, and the field equations reduce to the Weyl sector alone, with corrections suppressed by $(\Lambda/M_{\mathrm{Pl}})^2$ for astrophysical objects.

7 Linearized limit and modified Newtonian potential

We briefly summarize the linearized results (derived in detail in Paper II [9]).

7.1 Modified propagator

On flat Euclidean background, the graviton propagator in the transverse-traceless sector is

$$\tilde{G}_{\mathrm{TT}}(k^2) = \frac{1}{k^2\,\Pi_{\mathrm{TT}}(k^2/\Lambda^2)}, \tag{24}$$

with Π_{TT} given by (21). Factoring the denominator in the local (small-z) approximation gives poles at $k^2 = 0$ (massless graviton) and $k^2 = -m_2^2$ (massive spin-2), with

$$m_2^2 = \frac{60}{13}\,\Lambda^2, \qquad m_2 = \Lambda\sqrt{\frac{60}{13}} \approx 2.148\,\Lambda. \tag{25}$$

The scalar sector has an analogous mass

$$m_0^2(\xi) = \frac{\Lambda^2}{6(\xi - 1/6)^2}, \qquad m_0\big|_{\xi=0} = \sqrt{6}\,\Lambda \approx 2.449\,\Lambda, \tag{26}$$

which diverges at conformal coupling $\xi = 1/6$ (scalar decoupling).

7.2 Modified Newtonian potential

In the local Yukawa approximation, the modified gravitational potential is

$$\frac{V(r)}{V_{\mathrm{N}}(r)} = 1 - \frac{4}{3}\,e^{-m_2 r} + \frac{1}{3}\,e^{-m_0 r}, \tag{27}$$

where $V_{\mathrm{N}}(r) = -GM/r$ is the Newtonian potential. The Yukawa amplitudes $-4/3$ and $+1/3$ are parameter-free, determined entirely by the spin decomposition of the graviton propagator. At short distances:

$$\frac{V(r)}{V_{\mathrm{N}}(r)} \to 1 - \frac{4}{3} + \frac{1}{3} = 0 \quad \text{as } r \to 0, \tag{28}$$

so the $1/r$ singularity is regularized: $V(0)$ is finite. At large distances, both Yukawa terms decay exponentially and Newton's law is recovered.

7.3 Observational bounds

Confrontation with seven gravitational experiments (torsion-balance, time-delay, Casimir, satellite, and lunar-ranging measurements) yields the bound [9]

$$\Lambda > 2.565 \times 10^{-3}\ \mathrm{eV} \qquad (1/\Lambda < 77\ \mu\mathrm{m}), \tag{29}$$

at 95% confidence level. All post-Newtonian parameters satisfy experimental constraints with exponential margin [16].

8 FLRW cosmology

We now reduce the nonlinear field equations (16) to the Friedmann–Lemaître–Robertson–Walker metric, switching to Lorentzian signature.

8.1 FLRW geometry

The spatially flat FLRW metric is

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\,\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j. \tag{30}$$

The curvature quantities are

$$R = 6(\dot{H} + 2H^2), \tag{31}$$
$$R_{00} = -3(\dot{H} + H^2), \tag{32}$$
$$R_{ij} = (\dot{H} + 3H^2)\,a^2\delta_{ij}, \tag{33}$$
$$G_{00} = 3H^2, \tag{34}$$
$$G_{ij} = -(2\dot{H} + 3H^2)\,a^2\delta_{ij}, \tag{35}$$

where $H \equiv \dot{a}/a$ is the Hubble parameter.

8.2 Conformal flatness and Weyl sector elimination

Proposition 8.1. *On any FLRW background, $C_{\mu\nu\rho\sigma} = 0$.*

Proof. Every FLRW metric is conformally flat: $g_{\mu\nu} = \Omega^2(\eta)\,\eta_{\mu\nu}$ in conformal time. The Weyl tensor is conformally invariant in $d = 4$, and $C_{\mu\nu\rho\sigma}[\eta] = 0$ for the flat metric. □

This is the central simplification for cosmology. Since $C_{\mu\nu\rho\sigma} = 0$:

- The Bach tensor vanishes: $B_{\mu\nu} = 0$.
- The Weyl alpha-insertion vanishes: $\Theta^{(C)}_{\mu\nu} = 0$.

The entire Weyl sector, governed by $\alpha_C = 13/120$, drops out of the background cosmological equations. Only the R^2 sector contributes.

8.3 Reduced field equations on FLRW

On FLRW, the field equations reduce to

$$\frac{1}{\kappa^2}\,G_{\mu\nu} + \beta_R\left[F_2(\Box/\Lambda^2, \xi)\,H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\right] = \frac{1}{2}\,T_{\mu\nu}, \tag{36}$$

where $\beta_R \equiv \alpha_R(\xi)/(16\pi^2) = 2(\xi - 1/6)^2/(16\pi^2)$.

8.4 The $H_{\mu\nu}$ tensor on FLRW

Proposition 8.2. *On FLRW:*

$$H_{00} = -6H\dot{R} + \tfrac{1}{2}R^2 - 6R(\dot{H} + H^2), \tag{37}$$
$$H_{ij} = \left[2\ddot{R} + 4H\dot{R} - \tfrac{1}{2}R^2 + 2R(\dot{H} + 3H^2)\right]a^2\delta_{ij}, \tag{38}$$

with trace $g^{\mu\nu}H_{\mu\nu} = -6\,\Box R$ and $H_{\mu\nu}|_{\mathrm{dS}} = 0$.

Proof. From $\nabla_0\nabla_0 R = \ddot{R}$, $g_{00} = -1$, $\Box R = -\ddot{R} - 3H\dot{R}$, and $R_{00} = -3(\dot{H} + H^2)$:

$$\begin{aligned} H_{00} &= 2\ddot{R} + 2\Box R + \tfrac{1}{2}R^2 - 6R(\dot{H} + H^2) \\ &= 2\ddot{R} + 2(-\ddot{R} - 3H\dot{R}) + \tfrac{1}{2}R^2 - 6R(\dot{H} + H^2) \\ &= -6H\dot{R} + \tfrac{1}{2}R^2 - 6R(\dot{H} + H^2). \end{aligned}$$

The de Sitter limit follows: $\dot{H} = 0$ gives $R = 12H_0^2$, $\dot{R} = 0$, and $\frac{1}{2}(12H_0^2)^2 - 6(12H_0^2)(H_0^2) = 72H_0^4 - 72H_0^4 = 0$. □

8.5 Alpha-insertion on FLRW

On FLRW, all $\Box^k R$ are functions of t only by spatial homogeneity. Therefore $\nabla_\mu(\Box^k R) = \dot{R}_k\,\delta^0_\mu$, where $\dot{R}_k \equiv \frac{\mathrm{d}}{\mathrm{d}t}(\Box^k R)$.

Proposition 8.3. *The alpha-insertion components on FLRW are*

$$\Theta^{(R)}_{00} = -\frac{1}{2}\,S, \qquad \Theta^{(R)}_{ij} = -\frac{1}{2}\,S\,a^2\delta_{ij}, \tag{39}$$

where the source function is

$$S \equiv \sum_{n=1}^{N} \frac{f_{2,n}}{\Lambda^{2n}} \sum_{k=0}^{n-1} \dot{R}_k\,\dot{R}_{n-1-k}. \tag{40}$$

The equation of state of $\Theta^{(R)}$ is

$$w_\Theta \equiv \frac{p_\Theta}{\rho_\Theta} = +1 \qquad \text{(stiff matter)}, \tag{41}$$

since $\Theta^{(R)}_{00} = \Theta^{(R)}_{ij\,\mathrm{coeff}}$ (equal effective energy density and pressure). Both vanish on de Sitter ($\dot{R} = 0$).

8.6 Modified Friedmann equations

Taking the 00-component of (36) with $T_{00} = \rho$:

Theorem 8.4 (Modified first Friedmann equation)**.**

$$\boxed{H^2 = \frac{\kappa^2}{3}\,\rho - \frac{\kappa^2}{3}\,\beta_R\left[H_{00} + \Theta^{(R)}_{00}\right].} \tag{42}$$

The ij-component gives

$$\dot{H} + H^2 = -\frac{\kappa^2}{6}\,(\rho + 3p) + \frac{\kappa^2}{2}\,\beta_R\left[H_{ij\,\mathrm{coeff}} + \Theta^{(R)}_{ij\,\mathrm{coeff}}\right]. \tag{43}$$

8.7 Key properties of the modified Friedmann equations

1. **Conformal coupling ($\xi = 1/6$):** $\alpha_R = 0$, $\beta_R = 0$; all corrections vanish identically. Standard Friedmann equations are recovered.
2. **De Sitter:** $H_{00} = H_{ij} = 0$, $\Theta^{(R)} = 0$. Standard Friedmann: $3H^2 = \kappa^2\rho$.
3. **Late-time corrections:** For $H_0 \sim 10^{-33}$ eV and $\Lambda \sim M_{\mathrm{Pl}}$, the correction is $\mathcal{O}(\beta_R \cdot H_0^2/\Lambda^2) \sim 10^{-120}$.

8.8 Scalar perturbation equation

The scalar (Bardeen) perturbation Φ on the FLRW background obeys a modified Poisson equation. From the linearized analysis of Paper II [9], the gravitational potential in Fourier space satisfies

$$k^2\,\Phi = -4\pi G\,\frac{a^2\rho\,\delta}{\Pi_{\mathrm{s}}(k^2/\Lambda^2,\,\xi)}, \tag{44}$$

where δ is the matter density contrast and $\Pi_{\rm s}$ is the scalar propagator denominator (21). Since $\Pi_{\rm s}(z,\xi) = 1 + 6(\xi - 1/6)^2 z\,\hat{F}_2(z)$ and $z = k^2/\Lambda^2$ is negligibly small at cosmological and structure-formation scales ($k/\Lambda \lesssim 10^{-28}$ even at the PPN bound), we have $\Pi_{\rm s} \approx 1$ and the standard Poisson equation is recovered.

At conformal coupling $\xi = 1/6$, the denominator is $\Pi_{\rm s} = 1$ *exactly* for all k: the scalar graviton mode decouples from the gravitational potential regardless of the momentum scale. This is the same decoupling observed in the linearized propagator [5] and in the FLRW background equations.

The growth equation for the density contrast $\delta(t)$ in the sub-Hubble limit is

$$\ddot{\delta} + 2H\dot{\delta} - \frac{4\pi G_{\rm eff}(k)}{a^3}\,\rho_m\,\delta = 0, \tag{45}$$

with $G_{\rm eff}(k) = G/\Pi_{\rm s}(k^2/\Lambda^2,\xi)$. The modification of the effective gravitational coupling is

$$\frac{G_{\rm eff}}{G} - 1 \approx -6\Big(\xi - \frac{1}{6}\Big)^2 \frac{k^2}{\Lambda^2}\,\hat{F}_2\Big(\frac{k^2}{\Lambda^2}\Big), \tag{46}$$

which at $k = 0.1\,h$ Mpc^{-1} and $\Lambda = \Lambda_{\rm PPN}$ evaluates to $\sim 5.5\times10^{-57}$—completely negligible compared to the percent-level sensitivity of current surveys.

9 Gravitational-wave speed

Theorem 9.1 (GW speed in SCT)**.** *On an FLRW background, the gravitational-wave speed satisfies*

$$c_T = c\,\big[1 + \mathcal{O}(H^2/\Lambda^2)\big]. \tag{47}$$

Proof. The argument has four steps.

Step 1. On FLRW, $C_{\mu\nu\rho\sigma} = 0$ at the background level. Transverse-traceless tensor perturbations $h_{ij}^{\rm TT}$ do generate nonzero $\delta C_{\mu\nu\rho\sigma}$, but the R^2 sector couples only to scalar perturbations via SVT decomposition [17]. Therefore only the Weyl sector affects GW propagation.

Step 2. On a Minkowski background, Lorentz invariance ensures $\omega^2 = k^2$. The $\Pi_{\rm TT}$ form factor modifies the *amplitude* of the propagator (the residue at the pole) but not the dispersion relation.

Step 3. On a curved FLRW background, corrections to the dispersion relation arise at $\mathcal{O}(H^2/\Lambda^2)$, as the form factor couples to background curvature.

Step 4. Numerically, for cosmological parameters ($H_0 = 2.2\times10^{-18}$ Hz, $\Lambda \geq 2.38\times10^{-3}$ eV):

$$|c_T/c - 1| \lesssim 10^{-62}, \tag{48}$$

which satisfies the GW170817 bound [18] $|c_T/c - 1| < 5.6\times10^{-16}$ by **46 orders of magnitude**. □

The parametric estimate for the deviation is

$$\frac{|c_T - c|}{c} \sim \frac{\alpha_C}{2}\Big(\frac{H}{\Lambda}\Big)^2. \tag{49}$$

Table 1 shows the result across the cutoff range.

Table 1: Gravitational-wave speed deviation $|c_T/c - 1|$ for various values of the spectral cutoff Λ. The GW170817 bound is 5.6×10^{-16}.

Λ (eV)	$(H_0/\Lambda)^2$	$\lvert c_T/c - 1\rvert$	Satisfies GW170817?
2.38×10^{-3}	4.0×10^{-61}	2.2×10^{-62}	Yes
1	2.3×10^{-66}	1.2×10^{-67}	Yes
10^{10}	2.3×10^{-86}	1.2×10^{-87}	Yes
10^{19}	2.3×10^{-104}	1.2×10^{-105}	Yes

10 De Sitter stability

Proposition 10.1 (De Sitter is an exact, stable solution)**.** *On de Sitter ($H = H_0 = \text{const}$, $\dot{H} = 0$):*

$$H_{\mu\nu}\big|_{\text{dS}} = 0, \qquad \Theta^{(R)}_{\mu\nu}\big|_{\text{dS}} = 0. \tag{50}$$

The modified Friedmann equations reduce to standard GR: $3H_0^2 = \kappa^2\rho$.

For perturbations $H(t) = H_0 + \epsilon(t)$ with $|\epsilon| \ll H_0$, the decay rate is

$$\Gamma = 3H_0\big(1 + \mathcal{O}(\beta_R \cdot H_0^2/\Lambda^2)\big). \tag{51}$$

De Sitter is a stable attractor: perturbations decay exponentially.

Proof. The exactness follows from $\dot{H} = 0$, which implies $R = 12H_0^2 = \text{const}$, $\dot{R} = 0$, $\Box R = 0$. All terms in $H_{\mu\nu}$ and $\Theta^{(R)}_{\mu\nu}$ vanish.

For stability, the effective masses

$$m_2 = \Lambda\sqrt{\frac{60}{13}} \approx 2.148\,\Lambda, \qquad m_0(\xi{=}0) = \sqrt{6}\,\Lambda \approx 2.449\,\Lambda, \tag{52}$$

both satisfy $m \gg \frac{3}{2}H_0$ for all cosmological parameters (Higuchi bound), confirming that neither the spin-2 nor spin-0 massive mode becomes tachyonic near de Sitter. □

11 Late-time cosmological suppression

11.1 Central result

The fractional correction to the Friedmann equation is

$$\frac{\delta H^2}{H^2} = \beta_R(\xi)\left(\frac{H(z)}{\Lambda}\right)^2, \tag{53}$$

where $H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$. At the PPN-1 lower bound $\Lambda = 2.38 \times 10^{-3}$ eV, with $\xi = 0$ and $z = 0$:

$$\boxed{\left.\frac{\delta H^2}{H^2}\right|_{z=0} = 1.28 \times 10^{-64}.} \tag{54}$$

This is **64 orders of magnitude** below the ~ 0.18 modification needed to resolve the Hubble tension [19].

11.2 Derivation of the numerical value

The computation proceeds as follows:

1. $\alpha_R(0) = 2(0 - 1/6)^2 = 1/18 = 0.05556$.
2. $\beta_R(0) = (1/18)/(16\pi^2) = 3.518 \times 10^{-4}$.
3. $H_0 = 67.36$ km/s/Mpc $= 1.437 \times 10^{-33}$ eV.
4. $(H_0/\Lambda)^2 = (1.437 \times 10^{-33}/2.38 \times 10^{-3})^2 = 3.645 \times 10^{-61}$.
5. $\delta H^2/H^2 = 3.518 \times 10^{-4} \times 3.645 \times 10^{-61} = 1.282 \times 10^{-64}$.

11.3 Suppression across parameter space

Table 2: Fractional Hubble correction $\delta H^2/H^2$ across the parameter space.

Configuration	$\delta H^2/H^2$	$\log_{10}$	Comment
$z = 0$, $\Lambda = 2.38 \times 10^{-3}$ eV, $\xi = 0$	1.28×10^{-64}	-63.9	PPN bound
$z = 0$, $\Lambda = 1$ eV, $\xi = 0$	7.26×10^{-70}	-69.1	
$z = 0$, $\Lambda = 1$ GeV, $\xi = 0$	7.26×10^{-88}	-87.1	
$z = 0$, $\Lambda = M_{\rm Pl}$, $\xi = 0$	$\sim 10^{-125}$	-125.3	
$z = 1100$, $\Lambda = \Lambda_{\rm PPN}$, $\xi = 0$	5.40×10^{-56}	-55.3	CMB epoch
$z = 0$, $\Lambda = \Lambda_{\rm PPN}$, $\xi = 1/6$	0 (exact)	—	Conformal

Even at the CMB epoch ($z = 1100$), the largest correction is $\sim 10^{-56}$, still 55 orders below the required ~ 0.18.

11.4 Effective equation of state

The spectral correction acts as an effective fluid with

$$w_{\rm eff} = -1 + \delta w, \qquad \delta w = 2\,\beta_R \left(\frac{H}{\Lambda}\right)^2 = 2.6 \times 10^{-64} \tag{55}$$

at $z = 0$, $\xi = 0$, $\Lambda = \Lambda_{\rm PPN}$. The effective equation of state is $w_{\rm eff} = -1$ to 63 decimal places.

11.5 Growth function and S_8

From the modified Poisson equation, the gravitational coupling is $G_{\rm eff}/G_{\rm N} = 1/\Pi_{\rm s}(k^2/\Lambda^2, \xi)$. At structure-formation scales $k \sim 0.1\,h$ Mpc^{-1} ($k \approx 4.3 \times 10^{-31}$ eV):

$$\frac{\delta G_{\rm eff}}{G_{\rm N}} \sim 5.5 \times 10^{-57}, \tag{56}$$

which is 55 orders below the $\sim 9\%$ modification needed to resolve the S_8 tension.

11.6 Physical interpretation

The suppression $\delta H^2/H^2 \propto (H/\Lambda)^2$ is a *structural* property of any UV-complete theory with a spectral cutoff $\Lambda \gg H_0$. This is not specific to the particular SCT form factors; it holds for any theory where the one-loop effective action involves entire functions of $\Box/\Lambda^2$. Perturbative loop corrections cannot generate the $\Box^{-1}$ operators needed for infrared nonlocality [20]. Consequently:

- SCT is trivially consistent with all precision late-time cosmological data (Planck [19], DESI [21], Pantheon+, KiDS, DES, GW170817).
- SCT cannot resolve the H_0 or S_8 tensions. These would require infrared nonlocality with mass scale $m \sim H_0$.
- The testable predictions of SCT lie at shorter distance scales: solar-system tests [9], laboratory experiments, and early-universe physics.

12 Comparison with other approaches

12.1 Stelle gravity

The local ($\Lambda \to \infty$) limit of the SCT field equations (16) recovers Stelle's higher-derivative gravity [15, 12], with specific SM-determined coefficients (20). Stelle gravity is renormalizable but contains a massive spin-2 ghost at $k^2 = -m_2^2$. In SCT, the nonlocal form factor $\Pi_{\mathrm{TT}}(z)$ modifies the pole structure: the propagator has additional complex zeros [5, 22], and the treatment of the ghost degree of freedom requires the fakeon prescription [23] or an analogous mechanism.

The Stelle trace equation $-R - 6\beta\,\Box R = \kappa^2 T$ generalizes to the nonlocal trace (23), which includes the full form factor $F_2(\Box/\Lambda^2)$ rather than a constant coefficient. This introduces a tower of higher-derivative corrections that soften the UV behavior.

12.2 $f(R)$ gravity

The $f(R)$ theories of gravity [24, 25] are local theories with a modified Einstein–Hilbert action $S = \int \mathrm{d}^4x\,\sqrt{g}\,f(R)/(2\kappa^2)$. The most studied case, $f(R) = R + R^2/(6M^2)$ (Starobinsky [26]), drives inflation through a scalar degree of freedom (the scalaron) with mass M.

In the SCT framework, the R^2 sector of the action (3) produces a structurally similar modification, with two differences: (i) the coefficient $\alpha_R(\xi)$ is calculable from the SM particle content rather than being a free parameter; and (ii) the form factor $F_2(\Box/\Lambda^2)$ introduces nonlocal corrections beyond the leading R^2 term. At conformal coupling $\xi = 1/6$, $\alpha_R = 0$ and the R^2 sector decouples entirely; this is a distinctive prediction absent from generic $f(R)$ models.

12.3 Nonlocal gravity

Infinite derivative gravity (IDG) theories [27, 28, 22] share the most structural similarity with SCT: they employ entire form factors of the form $e^{-\Box/M_s^2}$ to regularize the graviton propagator in the UV. Both SCT and IDG produce cosmological corrections suppressed by $(H/M_s)^2 \ll 1$ when $M_s \gg H_0$, ensuring automatic consistency with late-time data.

The key differences are:

- **Origin:** SCT form factors arise from a one-loop heat kernel computation for the spectral action of a specific noncommutative geometry; IDG form factors are postulated to achieve desired UV properties.
- **Uniqueness:** The SCT form factors, once the SM particle content and the spectral action principle are fixed, are unique up to the single parameter ξ. IDG allows greater freedom in choosing the form factor.
- **Coefficients:** The local limits $\alpha_C = 13/120$ and $\alpha_R(\xi)$ are calculable in SCT, while in IDG the corresponding parameters are free.

Infrared nonlocal models (Maggiore's RT model [20], Deser–Woodard [29]) employ $\Box^{-1}$ operators with a mass scale $m \sim H_0$. These can partially address the H_0 tension ($\delta w \sim 0.14$) but face challenges with UV behavior and quantum stability. Table 3 summarizes the comparison.

Table 3: Comparison of late-time cosmological corrections across modified gravity approaches.

Model	Nonlocality	Scale	δw	Resolves H_0?
SCT	UV: $F(\Box/\Lambda^2)$	$\Lambda \gg H_0$	10^{-64}	No
IDG	UV: $e^{-\Box/M_s^2}$	$M_s \gg H_0$	$\sim 10^{-64}$	No
Maggiore RT	IR: $m^2 R \Box^{-2} R$	$m \sim H_0$	0.14	Partially
Deser–Woodard	IR: $f(\Box^{-1} R)$	$m \sim H_0$	$\mathcal{O}(0.1)$	Partially
Stelle	Local	—	0	No
$f(R)$	Local	M	Model-dep.	If tuned

12.4 Structural suppression theorem

The pattern visible in Table 3 reflects a general result:

Proposition 12.1 (UV suppression of cosmological corrections)**.** *Any gravitational effective action of the form*

$$S = \frac{1}{2\kappa^2} \int \mathrm{d}^4x \sqrt{g} \left[R + \alpha \, \mathcal{R} \, F(\Box/M^2) \, \mathcal{R} \right], \tag{57}$$

where $\mathcal{R}$ denotes any curvature scalar or tensor, $F(z) = \sum c_n z^n$ is an entire function with $F(0)$ finite, and $M \gg H_0$, produces corrections to the Friedmann equation that scale as

$$\frac{\delta H^2}{H^2} \sim \alpha \, F(0) \left(\frac{H}{M} \right)^2 \ll 1. \tag{58}$$

This follows directly from dimensional analysis: on FLRW, the curvature scale is set by H, and any nonlocal correction from $F(\Box/M^2)$ can at most introduce polynomial dependence on H^2/M^2. The entireness of F prevents $\Box^{-1}$ terms that would introduce infrared-enhanced contributions.

The implication is clear: UV-complete nonlocal gravity theories (SCT, IDG) are structurally incapable of producing order-unity cosmological corrections unless $M \sim H_0$, which would conflict with short-distance tests of gravity. Conversely, the automatic consistency of such theories with cosmological data is not an achievement but a structural inevitability.

12.5 Black hole entropy

The Wald entropy formula [30] applied to the curvature-squared Lagrangian (3) gives the black hole entropy correction

$$S_{\mathrm{BH}} = \frac{A}{4G} + \frac{1}{16\pi^2} \oint_{\mathcal{H}} \left[2F_1(0) \, \frac{\partial C^2}{\partial R_{\mu\nu\rho\sigma}} + F_2(0,\xi) \, \frac{\partial R^2}{\partial R_{\mu\nu\rho\sigma}} \right] \hat{\epsilon}_{\mu\nu} \hat{\epsilon}_{\rho\sigma} \, \mathrm{d}A, \tag{59}$$

where $\hat{\epsilon}_{\mu\nu}$ is the binormal to the bifurcation surface. In the local limit (on the horizon, where the form factors reduce to their $z = 0$ values), the dominant term is the Bekenstein–Hawking entropy $A/(4G)$, with higher-derivative corrections of order $\alpha_C\,(r_s/\ell_\Lambda)^2$ and $\alpha_R(\xi)\,(r_s/\ell_\Lambda)^2$, where r_s is the Schwarzschild radius and $\ell_\Lambda = 1/\Lambda$.

For astrophysical black holes ($r_s \gg \ell_\Lambda$), the corrections are completely negligible. They become significant only when $r_s \sim \ell_\Lambda$, i.e., for hypothetical black holes with mass $M_{\rm BH} \sim M_{\rm Pl}^2/\Lambda$. This regime is relevant for the endpoint of black hole evaporation and for Planck-scale physics.

13 Conclusions

We have derived the full nonlinear variational field equations of the one-loop spectral action with Standard Model content on arbitrary curved backgrounds. The field equations (16) extend the Einstein equations with three types of corrections: the nonlocal Bach tensor $B_{\mu\nu}$ (from the Weyl-squared sector), the tensor $H_{\mu\nu}$ (from the R^2 sector), and the Barvinsky–Vilkovisky alpha-insertion tensors $\Theta_{\mu\nu}^{(C,R)}$ (from the metric dependence of $\Box$).

The field equations satisfy six consistency conditions: (i) the local limit recovers Stelle's higher-derivative gravity with calculable SM coefficients $\alpha = 13/(120 \cdot 16\pi^2)$ and $\beta = 2(\xi - 1/6)^2/(16\pi^2)$; (ii) the linearized limit reproduces the modified propagator and Newtonian potential of Paper II; (iii) the Bianchi identity is satisfied; (iv) the trace structure is correct; (v) de Sitter spacetime is an exact solution; (vi) Ricci-flat backgrounds simplify to the Weyl sector.

The reduction to FLRW cosmology exploits the conformal flatness of any FLRW metric, which eliminates the entire Weyl sector from the background equations. Only the R^2 sector, governed by $\alpha_R(\xi) = 2(\xi - 1/6)^2$, contributes to the modified Friedmann equations. The five central cosmological results are:

1. **Gravitational-wave speed:** $c_T = c$ to $\mathcal{O}(H^2/\Lambda^2)$, satisfying the GW170817 bound by 46 orders of magnitude.

2. **De Sitter stability:** De Sitter is an exact solution and a stable attractor. The effective masses $m_2 \approx 2.148\,\Lambda$ and $m_0 \approx 2.449\,\Lambda$ (at $\xi = 0$) satisfy the Higuchi bound.

3. **Conformal decoupling:** At $\xi = 1/6$, all spectral corrections to the Friedmann equations vanish identically.

4. **Late-time suppression:** $\delta H^2/H^2 = 1.28\times10^{-64}$ at the PPN lower bound, ensuring automatic consistency with all precision cosmological data.

5. **Structural limitation:** SCT cannot resolve the H_0 or S_8 tensions, which require infrared nonlocality at a scale $m \sim H_0$.

Together with the companion papers on form factors [5] and solar-system tests [9], the present results establish the phenomenological viability of the spectral action framework: it modifies gravity at the scale $r \sim 1/\Lambda$ while preserving general relativity at both solar-system and cosmological distances. The principal open questions concern the treatment of the massive spin-2 ghost mode (the fakeon prescription versus alternative quantization schemes), the all-orders UV behavior (perturbative finiteness beyond two loops), and the early-universe dynamics (inflation driven by the R^2 sector).

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A Form factor identities

We collect the key identities for the master function and form factors used throughout this paper.

A.1 Master function

The master function (7) has the Taylor expansion

$$\varphi(x) = \sum_{n=0}^{\infty} \frac{(-1)^n\, n!}{(2n+1)!}\, x^n, \tag{60}$$

with $\varphi(0) = 1$, $\varphi'(0) = -1/6$. It is an entire function of order 1/2, with the integral representation

$$\varphi(x) = \int_0^1 \mathrm{d}s\; e^{-x\, s(1-s)}. \tag{61}$$

A.2 Spin-specific form factors

The heat kernel form factors for each spin sector are:
Spin-0 (scalar):

$$h_C^{(0)}(x) = \frac{1}{12x} + \frac{\varphi(x)-1}{2x^2}, \qquad \beta_W^{(0)} = h_C^{(0)}(0) = \frac{1}{120}. \tag{62}$$

Spin-1/2 (Dirac):

$$h_C^{(1/2)}(x) = \frac{3\varphi(x)-1}{6x} + \frac{2(\varphi(x)-1)}{x^2}, \qquad \beta_W^{(1/2)} = \frac{1}{20}. \tag{63}$$

Spin-1 (vector):

$$h_C^{(1)}(x) = \frac{\varphi(x)}{4} + \frac{6\varphi(x)-5}{6x} + \frac{\varphi(x)-1}{x^2}, \qquad \beta_W^{(1)} = \frac{1}{10}. \tag{64}$$

A.3 Standard Model totals

With $N_s = 4$, $N_D = 45/2$, $N_v = 12$:

$$\alpha_C = N_s\, \beta_W^{(0)} + N_D\, \beta_W^{(1/2)} + N_v\, \beta_W^{(1)} = \frac{4}{120} + \frac{45}{40} + \frac{12}{10} = \frac{13}{120}, \tag{65}$$

$$\alpha_R(\xi) = 2\Big(\xi - \frac{1}{6}\Big)^2. \tag{66}$$

The ξ-independence of α_C is a consequence of the conformal invariance of the Weyl-squared action.

A.4 Normalized form factors at selected points

The propagator denominators evaluate to:

$$\Pi_{\mathrm{TT}}(0) = 1, \qquad \Pi_{\mathrm{TT}}(1) \approx 1.179, \qquad \lim_{z\to\infty} z\,\alpha_C\,\hat{F}_1(z) = -\frac{89}{12}. \tag{67}$$

The UV asymptotic $x\,\alpha_C\,\hat{F}_1(x) \to -89/12$ governs the large-momentum behavior of the graviton propagator.

A.5 Gauss–Bonnet consistency at the variational level

The variation of the three curvature-squared invariants satisfies the Gauss–Bonnet identity:

Tensor structure	$\delta(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})$	$4\times\delta(R_{\mu\nu}R^{\mu\nu})$	$\delta(R^2)$	Sum
$\Box R_{\mu\nu}$	4	4	0	0
$g_{\mu\nu}\Box R$	4	2	-2	0
$\nabla_\mu\nabla_\nu R$	-6	-4	2	0
$R_{\mu\rho\nu\sigma}R^{\rho\sigma}$	8	8	0	0
$g_{\mu\nu}R_{\mu\nu}R^{\mu\nu}$	-2	-2	0	0
$g_{\mu\nu}R^2$	$1/2$	0	$-1/2$	0
$R\,R_{\mu\nu}$	-2	0	2	0

All seven sums vanish, confirming the Gauss–Bonnet identity $\delta(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) - 4\,\delta(R_{\mu\nu}R^{\mu\nu}) + \delta(R^2) = 0$ at the variational level. This identity was used to verify the dual derivation of the field equations (Sec. 5).

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